AIRSTREET · TRACEABILITY MATRIX

AirStreet traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-30	Amish Chadha	Generated by .kit/trace.py
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AirStreet traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 19 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	PM2.5 at street level	AST-CAL-001, AST-STM-001, AST-OPT-001	Analysis (AST-CAL-001 E), TRL 3; CalRig chamber check and field collocation, TRL 4	At risk: within the bound up to 80 % RH; outside it at 85 % RH and above, where water uptake makes the correction uncertain
UR-002	PR-002	NO2 at street level	AST-CAL-001, AST-STM-001, AST-OPT-001	Analysis (AST-CAL-001 C), TRL 3; collocation with a chemiluminescence reference, TRL 4	At risk: about 4.3 ppb from a budget of literature and assumed terms, with the datasheet noise over 12 independent records an hour and a 5 % slope term (AST-DDR-005 F1 and the independent review); hours outside the sensor's rated range are flagged (PR-007)
UR-003	PR-003	NO2 near the WHO annual guideline	AST-CAL-001	Collocation statistics (research)	Withdrawn: annual uncertainty about 4.65 ug/m3 (1 sigma); no WHO guideline claim
UR-003	PR-004	Streets above the EU NO2 limit value	AST-CAL-001, AST-OPT-001	Analysis (AST-CAL-001 C), TRL 3; collocation statistics, TRL 4	At risk: decisive only below 33.9 or above 46.1 ug/m3 against 40, and below 13.9 or above 26.1 ug/m3 against 20 (at the directive's 20 degC; AST-CAL-001 [C9]). Both limits since Amish's decision 4 of 2026-10-30 (AST-DDR-006); the recast directive sets 20 ug/m3 from 2030 (AST-REG-001 A2)
UR-001, UR-002	PR-005	Temperature and humidity for corrections	AST-CAL-001, AST-BLD-001	Analysis (AST-CAL-001 D), TRL 3; CalRig and side-by-side with an aspirated reference, TRL 4	At risk: SHT45 meets the sensor figures; shield error 0.42 to 0.87 K at 1 m/s, which reads 4.1 % RH low at 85 % RH
UR-005	PR-006	Stated calibration method	AST-CAL-001	Inspection of the calibration record per node, TRL 4	Met by design (AST-DDR-001 D3)
UR-006	PR-007	Raw data for recalibration	AST-CAL-001, AST-BLD-001	Inspection of the firmware and data schema, TRL 4	Met by design: flags set when the shield air is above 35 degC, RH above 80 % (PM) or 85 % (NO2) or below 20 % (PM) or 15 % (NO2), or below 10 degC (PM)
UR-004	PR-008	Reporting for street maps	AST-CAL-001, AST-BLD-001	Analysis of timing, airtime and storage (AST-CAL-001 B), TRL 3	Met on paper: at 15 min TTN fair use is met up to SF9; 7 days in 40.3 kB

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-013	PR-009	FieldNode power budget	AST-CAL-001, AST-BLD-001	Power budget (AST-CAL-001 A), TRL 3; energy log, TRL 4	Met on paper: 87.0 mW design load; 2.33 Wh/day against 7.75 Wh stored in the worst month; 6.6 days without sun
UR-016	PR-010	Radio duty cycle	AST-CAL-001	Airtime calculation (AST-CAL-001 B), TRL 3	Met on paper: 0.082 % at SF9, 0.60 % at SF12; 247 ms at SF9 in US915
UR-007	PR-011	Pole mounting without drilling	AST-CAL-001, AST-BLD-001	Design review and model (AST-CAL-001 F), TRL 3	Met by design: inlet 3.0 m (AST-DDR-001 D7); 140 degree V-saddles and bands cut to length
UR-009	PR-012	Survive outdoors	AST-CAL-001, AST-BLD-001, AST-STM-001	Design review (AST-CAL-001 H), TRL 3; field trial, TRL 4	At risk: the NO2 cell is rated to 40 degC and 85 % RH and the SPS30 is recommended to 40 degC and 80 % RH, so hours beyond are flagged (PR-007); condensation and insects unverified
UR-008	PR-013	Light and compact	AST-CAL-001, AST-BLD-001	Constructable model (AST-CAL-001 F), TRL 3; weighing, TRL 4	Met on paper: 3.86 kg with the sun shield (3.70 kg without), 0.14 kg margin; 0.144 m2, 0.006 m2 under the limit, with the tube hood of AST-DDR-006; the pole owner's confirmation of the load is still to come
UR-011	PR-014	Levels only	AST-CAL-001	Design review, TRL 3	Met by design
UR-012	PR-015	Low cost	AST-CAL-001	Priced BOM (AST-CAL-001 I), TRL 3	Over the value-engineering target by $21.00**: sensor head $306.00 (the tube hood and its fixings add $3.50); $437.00 with the FieldNode core
UR-010	PR-016	Serviceable in the field	AST-CAL-001, AST-BLD-001	Service time estimate (AST-CAL-001 H), TRL 3; timed exchange, TRL 4	Met on paper: about 14 min for a pod exchange
UR-014	PR-017	Open and garage buildable	AST-BLD-001	Inspection of the build plan AST-BLD-001 and the drawings, TRL 3	Met by design (text accepted by Amish on 2026-10-30, AST-DDR-006)
UR-015	PR-018	Installation and service safety	AST-BLD-001	Inspection of the build plan safety stops and AST-RSK-001, TRL 3	Met by design (text accepted by Amish on 2026-10-30, AST-DDR-006)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-003	PR-019	Independent check of annual NO2	AST-BLD-001	Inspection of the model, BOM line 12 and drawing AST-DWG-110, TRL 3; tube results against the node's annual mean, TRL 4	Met by design: hood and three clips in the model, 0.081 kg, $2.50 in the BOM; the tubes are bought with laboratory analysis (about £8 each, 2015 price) and are a running cost, not in the BOM (decision 2 of 2026-10-30, AST-DDR-006)

Gaps

None.